PATTERN_S_MARKOV: |S|² label chain stationary law → T-fibre vs MF2

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RESULTS: PATTERN_S_MARKOV (Map-PSM1)

PREREG sha256 cdbee31a0922f00a139996337f88e4bd02bf31c38750a84ae0da0972801b0bc8. Hashed after explore and before test.

fibre	ρ_res (MF2)	push(π), π = S-orbit average	class (0,0)	class (1,1)
0	1/32	8/35	16/35	0
1	1/4	11/70	0	11/35
2	3/32	0	0	0
3	0	9/70	0	9/35
4	7/32	19/70	19/35	0
5	1/4	3/35	0	6/35
6	5/32	0	0	0
7	0	9/70	0	9/35

- TV = 71/140.
- The push equals the PB2 vector exactly.
- The mixed class (0,1)+(1,0) pushes to the same vector as the full average.

Verdict: FAIL (Category A, exact). Bridge: **NO.** ## NOTES: PATTERN_S_MARKOV ## S matrix used - Taken directly from the written S-law (rev8 l.274): each label goes to 35 targets with |coef|² = 1/35. - The A35 / 105-companion S was not needed: the written label S-law is already a 140×140 mixer. - I verified 2a ∈ L′, so the β-index mod L′ is consistent.

Lemma PSM1 (Category A)

a. The |S|² chain is doubly stochastic. It is reducible into exactly 3 closed classes indexed by (coset α, coset β): (0,0) and (1,1) (35 labels each), and (0,1) ↔ (1,0) (70 labels, period 2). The stationary space has dimension 3, and each class's law is uniform.

b. Under the T-phase fibre, every stationary law (any convex mix of the 3 class laws) has zero mass on fibres 2 and 6. MF2 has 3/32 + 5/32 there. So **no** stationary law of the S-chain matches MF2.
c. The S-orbit average equals the uniform pool, so the S-dynamics adds nothing beyond PB2 on this fibre.

Observation (post hoc, not a claim)

- The (1,1) class (a-cosets) supports {1,3,5,7}, the odd fibres. The (0,0) class supports {0,4}.
- MF2 is not a mix of these, because it needs {2,6}.
- Under MF1, MF2's open fibres split as {1,5} odd-open and {0,2,4,6} even-open, with {3,7} closed. The Resonant T-fibre has an exactly opposite parity block structure.

Next pattern step

The written S/T action is now exhausted as label dynamics (PSM1). The Markov analogue that MF2 actually is, a chain on residues u mod 2^k, would need a written Resonant chain on Z/2^k or Z/8, such as the T-phase action of T on labels composed with S (the ST order-3 element). The cheapest written next test is the **ST chain**: the transition $|(ST)_{ij}|^2$ from the written T-law and S-law, whose stationary laws on the 3 classes may mix differently because T permutes β (β → α+β+w_c). If that also fails, the written modular side has no 8-fibre law matching MF2. The honest stopping point is then to report this to Arvin and ask for his intended Π/grading.